



DEPARTMENT OF APPLIED COMPUTING & ARTIFICIAL INTELLIGENCE

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SECP3106 APPLICATION DEVELOPMENT (WBL)

Section 01

INDUSTRY PROJECT PROPOSAL

in collaboration with

TVETMARA Johor Bahru (Institut Kemahiran Mara Johor Bahru)

Group CodeCraft

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1.0 Project Background

TVET MARA Johor Bahru (MARAJB) manages a significant academic population of 800 students overseen by 110 academic staff. To improve administrative efficiency, a centralized Web-Based Learning Management System (LMS) is required. Unlike traditional student-facing portals, this system is designed exclusively as a tool for staff and administrators to manage student records, attendance, and academic progression more effectively.

2.0 Objectives

The main objective of this project is to design and develop a comprehensive Attendance Management System that improves the efficiency, accuracy, and accessibility of attendance-related processes in an educational environment. The system aims to integrate multiple key functionalities into a single platform, including attendance recording, discipline issue reporting, report generation, timetable management, and class rescheduling. By enabling lecturers to mark attendance in real time, monitor student behaviours, and generate meaningful reports, the system supports better decision-making and student performance tracking. In addition, features such as timetable upload, timetable viewing, and replacement class booking are included to streamline scheduling and reduce administrative workload.

3.0 Target User

The proposed system is designed exclusively for the internal staff of TVET MARA Johor Bahru. While students are the subjects of the data, they do not have login access. The users are categorized into the following three primary roles:

3.1 Academic Lecturers (Primary Users)

- Lecturers remain the most critical user group as they perform the daily operational tasks. Under the new module structure, their responsibilities now include:

- **Attendance Execution:** Marking daily attendance (Present, Absent, MC, CK).
- **Schedule Management:** Viewing assigned timetable slots and booking replacement classes for missed sessions.
- **Behavioral Tracking:** Reporting discipline issues such as misconduct or frequent absenteeism directly through the system.

3.2 Program Coordinators & Administrators

- The shift toward "Uploading Timetables" and "Generating Reports" indicates a need for an administrative tier that oversees the department's logistics:
- **Logistics Management:** Responsible for uploading the master timetable data (classes, subjects, and assigned lecturers).
- **Resource Oversight:** Managing the "Booking Module" to ensure replacement classes do not clash with existing room or lecturer schedules.
- **Data Analysis:** Using the reporting module to view attendance statistics and trends across the entire program.

3.3 Student Affairs / Discipline Officers

- With the addition of a dedicated **M2: Report Discipline Issue** module, this specific user group becomes necessary for follow-up actions:
- **Intervention:** Monitoring the disciplinary reports filed by lecturers to identify students who fall below the 80% attendance threshold.
- **Documentation:** Reviewing the "Reporting Module" summaries to provide evidence for official warnings or dismissal processes.

4.0 Problem Statement

The current student attendance tracking and curriculum scheduling process of the TVETMARA campus in Johor Bahru relies heavily on manual operation and decentralised management, which leads to serious operational inefficiency. At present, student attendance

records are mainly obtained through time-consuming methods, and it is difficult to track long-term trends or identify frequent absences or violations in time. In addition, the lack of a unified digital system for lecturers to arrange courses and coordinate alternative courses has led to scheduling conflicts and administrative bottlenecks. There is no centralised platform to seamlessly upload curriculum, view daily class time and generate comprehensive attendance statistics, which hinders the ability of faculty and staff to effectively manage classroom affairs and monitor student participation.

5.0 Proposed Solution

In order to solve these challenges and ensure the feasibility of the project within the two-month development period, our team proposes to develop a centralised attendance system application. The application will simplify the entire attendance and scheduling process by integrating six professional sub-modules. The Attendance Record module will allow teachers to mark students' class grades and classify the status as presence, absence, or *cuti dengan kebenaran*. In order to solve the problem of behaviour tracking, the Report Discipline Issue module will record students' frequent absences or misconduct. The system will also have a Reporting module to display attendance statistics and summaries, including percentages and trends. For logistics and venue management, the Upload Time Schedule module will handle the upload of core course schedule data, including courses, subjects, instructors and times. Lecturers can use the Show Timetable Slot function to display their daily curriculum and directly access the attendance interface. Also, the Booking module will be implemented to make it easy for lecturers to reschedule courses or book alternative courses. By focussing all the effort on these interrelated submodules, the application will create a complete and modern system that will greatly reduce management costs and replace the outdated manual tracking processes.

6.0 Mandatory Module

6.1 Login System

The login system provides users with authentication methods to access the application. As the main security entry point, it will verify user credentials before granting access to the system. The system will also support role-based routing to ensure that students, instructors and administrators are automatically guided to their respective dashboards after successful authentication.

6.2 Registration System

The registration system facilitates the registration process for new users. This module will perform basic data verification to ensure that the format of the data entered such as student ID, employee number and password is correct before creating a new account in the registration database.

6.3 Manage User System

This User Management system is the central management centre for account management. This module enables authorised administrators to securely view, edit, or delete user accounts across the platform. In order to maintain strict data security and operational integrity when managing employee and student lists, the system will implement role-based access control. This ensures that management capabilities are appropriately limited based on user levels, such as students, instructors, and administrators, and only grant them permissions that match their specific roles.

7.0 Industry Module

7.1 Sistem Kedatangan Pelajar

As the first of six industry-specific modules, this module provides lecturers with a streamlined, checkbox-based interface for fast and intuitive attendance tracking. The system automatically translates these inputs into backend data, recording a "1" for present students and a "0" for

unexcused absences. For authorized leave, the system maintains a "1" to preserve the student's record but appends specific notes to distinguish the absence from regular attendance. Finally, the module features an automated reporting function that calculates and displays individual student attendance rates at the conclusion of the semester.

7.2 *Sistem Melapor Isu Disiplin*

This module is integrated with the attendance-taking system, where lecturers can directly report discipline issues by clicking a warning button beside a student's name during attendance marking. The system will then open a reporting form that allows lecturers to record detailed information to enable efficient tracking, trend analysis, and timely disciplinary or support actions.

7.3 *Sistem Laporan*

This module generates and displays student attendance statistics and summaries in a clear and organised way. It includes overall attendance percentage, attendance trends, and performance for each class to help lecturers monitor student participation. This module supports better decision-making by allowing lecturers to identify attendance patterns and take appropriate action when needed.

7.4 *Sistem Muat Naik Jadual Waktu*

This module is responsible for managing the uploading of timetable data into the system. Course codes, class groups, lecturers, classroom locations, and scheduled time slots are among the organized timetable details that authorized users, like administrators or heads of programs, can upload. Before putting the uploaded data in the central database, the system will verify its accuracy and consistency. This module guarantees that all academic scheduling data is consistent and readily available throughout the system by centralizing timetable data. It reduces errors brought on by out-of-date or conflicting timetable records and does away with the necessity for human schedule dissemination. Furthermore, this module is essential to the attendance system's operation. Accurate timetable data is necessary to guarantee that lecturers

can only record attendance for legitimate classes at the appropriate time because attendance marking is linked to class sessions. This connection guarantees that attendance records match the official schedule and increases data reliability.

7.5 *Sistem Paparan Slot Jadual Waktu*

This module allows lecturers to view their assigned timetable in a clear and structured format. Daily or weekly schedules including course information, times, and places will be shown by the system. Lecturers can more effectively organize their teaching activities and find future classes by simply navigating through their schedule. This module's direct interaction with the attendance system is one of its main features. Lecturers can quickly access the attendance interface for a particular class session by selecting it from the timetable that is displayed. This guarantees accurate and real-time attendance recording while cutting down on the time needed to find class information. This module enhances workflow efficiency, reduces scheduling confusion, and supports improved time management for lecturers by offering a consolidated and user-friendly timetable display. Additionally, it guarantees that every attendance record is connected to the appropriate class sessions, improving the overall consistency and dependability of the system.

7.6 *Sistem Tempahan Kelas Ganti*

The main purpose of this module is to enable lecturers or faculty administrators to easily adjust the course schedule or book alternative courses directly through the application. In the event of an unexpected situation such as sick leave, emergency meeting or public holiday, lecturers or via faculty administrators can use this function to quickly select the affected course time and provide an alternative schedule. In order to ensure a smooth workflow, this module will be used in combination with the uploaded course schedule data. This allows the system to automatically check students' availability and avoid scheduling problems or course conflicts. This module completely eliminates the time-consuming coordination work previously required between lecturers, students and management departments. Once the alternative course is successfully booked, the system will dynamically update the lecturer's schedule as well as the students' schedule so that the modified time period can be used for digital attendance records

immediately. This forms an efficient and conflict-free curriculum scheduling system and reducing interference with the teaching process.

8.0 Module Distribution

Module	Type	Lead Member
Login	Mandatory	Lau Yan Kai
Registration	Mandatory	Lau Yan Kai
Manage User	Mandatory	Lau Yan Kai
Student Attendance (<i>Kedatangan Pelajar</i>)	Industry	Syarifah Dania binti Syed Abu Bakar
Report Discipline Issues (<i>Melapor Isu Disiplin</i>)	Industry	Joanne Ching Yin Xuan
Reporting (<i>Laporan</i>)	Industry	Evelyn Goh Yuan Qi
Upload Timetable Schedule (<i>Muat Naik Jadual Waktu</i>)	Industry	Damiya Aina binti Basir Abd Shammad
Showing Timetable Slot (<i>Paparan Slot Jadual Waktu</i>)	Industry	Damiya Aina binti Basir Abd Shammad
Booking Module (<i>Tempahan Kelas Ganti</i>)	Industry	Lau Yan Kai

9.0 Justification

The necessity to solve inefficiencies in TVET MARA Johor Bahru's current manual and disjointed processes justifies the proposed Attendance Management System. Current procedures mostly rely on decentralized systems and paper-based records, which result in inconsistent data, delayed reporting, and challenges in efficiently monitoring student attendance and behaviour.

This system's modules have been carefully chosen to directly assist the main goal of enhancing attendance management. While the Discipline Reporting feature enables instructors to promptly record behavioral issues during class, the Student Attendance module offers real-time attendance recording. By producing attendance data and trends, the Reporting module improves decision-making even more.

A vital supporting function is also played by the schedule Management modules, which comprise schedule upload and timetable display. By ensuring that all attendance records are precisely connected to official class schedules, these modules help to avoid mistakes like registering attendance for the wrong sessions or time slots. Lecturers can more effectively record attendance and access their schedules by using a centralized, structured timetable system.

The system is further enhanced with the Booking Replacement Class module, which permits flexible class rescheduling without interfering with attendance tracking. This keeps correct records and guarantees continuity in academic activity.

Overall, the integration of these modules into a single platform reduces administrative workload, improves data accuracy, and enhances operational efficiency. The system also supports digital transformation by replacing outdated manual processes with a centralized, user-friendly solution that meets the institution's current academic management needs.

10.0 Conclusion

In conclusion, this project aims to develop a comprehensive Attendance Management System that integrates multiple modules to improve the efficiency and accuracy of attendance tracking and academic management processes. By providing features such as attendance recording, reporting, timetable management, and class rescheduling, the system will support lecturers and administrators in making better decisions and managing their tasks more effectively. By the end of the semester, our team aims to deliver a functional and user-friendly application that addresses the current challenges faced by the institution.